



HK IMO Training 2022-2023
Problem Set 5
For 29 October 2022



Problem 17. Let a, b, c be real numbers satisfying $|a - b| \geq |c|$, $|b - c| \geq |a|$ and $|c - a| \geq |b|$. Prove that one of a, b, c is equal to the sum of the other two.

Problem 18. Let $ABCDE$ be a cyclic pentagon such that $AE = DE$. The diagonals AC and BD meet at P . Let X be a point on the ray BA beyond A such that $AX = PD$. Let Y be a point on the ray CD beyond D such that $DY = PA$. Prove that PE is perpendicular to XY .

Problem 19. There are n trainees in the IMO training. In a phase, there are 7 classes in total. For every pair of trainees, there are at least 3 classes such that exactly one of them has attended the class (it is not necessary that the same trainee joins all those 3 classes). Also, it is known that one of the trainees has never joined any class in this phase. Find the greatest possible number of n .

Problem 20. Find all polynomials $P(x)$ with integer coefficients satisfying the following condition: For any positive integers m and n such that $m + n$ is a perfect square, $P(m) + P(n)$ is also a perfect square.